INCH–POUND

MIL–PRF–51526B
w/AMENDMENT 1
27 February 2006
SUPERSEDING
MIL-PRF-51526B
15 December 2004

PERFORMANCE SPECIFICATION

FILTER, PARTICULATE, 200 CFM

Reactivated for New Design after 13 March 2000

This specification is approved for use by the U.S. Army Edgewood Chemical Biological Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of High Efficiency Particulate Air (HEPA) filter rated at 200 standard cubic feet per minute (scfm) for use in a gas-particulate filter set (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD–ECB–ENA–S, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010–5424 or emailed to SpecsTeam@apgea.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

DEPARMENT OF DEFENSE SPECIFICATIONS

MIL−S−901	–	Shock Test, High−Impact Shipboard Machinery, Equipment, and Systems, Requirements for

DEPARTMENT OF DEFENSE STANDARDS

MIL−STD−130	–	Identification Marking of U.S. Military Property
MIL−STD−167−1	–	Mechanical Vibrations of Shipboard Equipment
MIL−STD−282	–	Military Standard, Filter Units, Protective Clothing, Gas−Mask Components and Related Products: Performance Test Methods
MIL−STD−810	–	Environmental Engineering Considerations and Laboratory Tests
MIL−STD−1168	–	Ammunition Lot Numbering and Ammunition Data Cards

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111−5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

NAVAL SEA COMMAND

Drawing 53711−6573660	–	Filter, Gas
Drawing 53711−6263500	–	Filter System, CPS
SS200−AG−MMM−010	–	Technical Manual for Navy Shipboard Collective Protection System (CPS), CBR Filter System Operation and Maintenance

(Copies are available from Naval Surface Warfare Center Dahlgren Division, 17320 Dahlgren Road, Code G52, Dahlgren, VA 22448−5100.)

2.3 Non−Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN CONFERENCE OF GOVERNMENTAL INDUSTRIAL HYGIENISTS

Publication – Threshold Limit Values and Biological Exposure Indices

(Copies are available from ACGIH, 1330 Kemper Meadow Dr., Suite 600, Cincinnati, OH 45240 or http://www.acgih.org.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 209 – Standard Specification for Aluminum and Aluminum Alloy Sheet and Plate

(Copies are available from www.astm.org or ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428–2959.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AMS 3195 – Silicone Rubber Sponge, Closed Cell, Medium

(Copies are available from SAE, 400 Commonwealth Dr., Warrendale, PA 15096 or http://www.sae.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 12103–1 – Road Vehicles – Test Dust for Filter Evaluation – Part 1

(Copies are available online from http://www.iso.org.)

UNDERWRITERS LABORATORY, INC (UL)

UL900 – Air Filter Units – Standard for Safety

(Copies are available online from http://www.comm–2000.com.)

2.4 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspections in accordance with 4.2.

3.2 Interface requirements.

3.2.1 Size and shape. Each particulate filter shall be cylindrical in form. It shall fit inside the gas filter (MIL–PRF–51525) and be removed without damage to itself or the gas filter. The filter shall conform to the interface dimensions in Figure 1. Each end of the filter or end cover shall contain one groove for gasket location. The gasket shall be grade number 2C1 of

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ASTM D 1056 or silicone rubber per SAE–AMS–3195, be seamless, and have a thickness of 0.31 inch. The gasket shall be permanently attached to the filter with an adhesive.

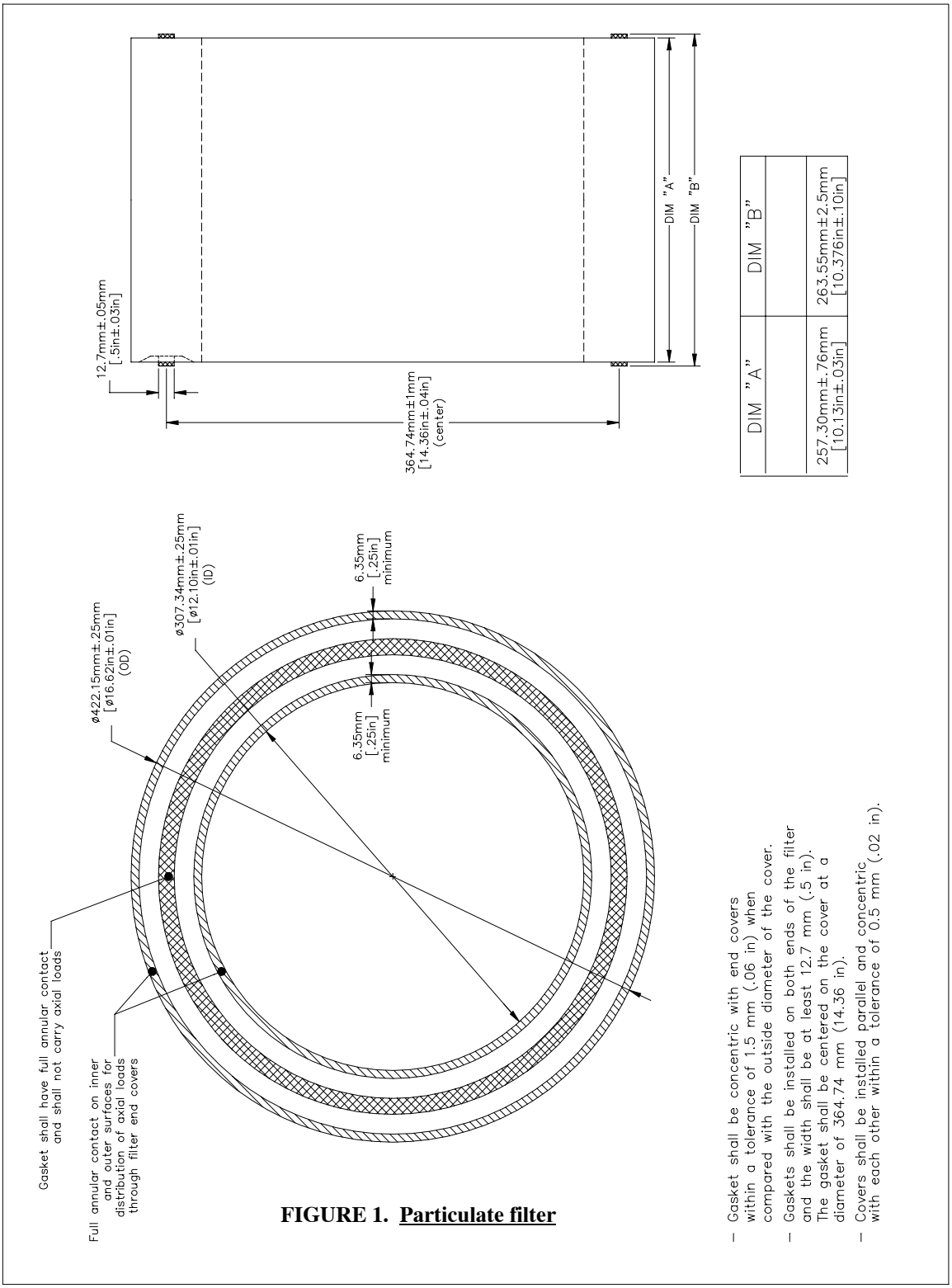


FIGURE 1. Particulate filter

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3.2.2 Weight. Each particulate filter shall weigh no more than 13 pounds.

3.2.3 Interchangeability. Each particulate filter and any associated gasket shall be individually interchangeable (replaceable) by one of similar form, fit and function without modification of the gas filter or filter housing.

3.3 Operating requirements.

3.3.1 Airflow resistance. The airflow resistance (pressure drop) of the filter shall not exceed 0.7 inches of water gage (iwg) at the rated airflow of 200 scfm. Standard conditions is defined as an airstream temperature of 70°F and a barometric pressure of 760 millimeters (mm) Hg.

3.3.2 Filtration efficiency. The particulate collection efficiency of 0.3−micron diameter particles shall be at least 99.97 percent when the filter is challenged, at the rated airflow, by an aerosol.

3.4 Environmental requirements.

3.4.1 Hot/cold temperature. After temperature conditioning at −50 and 160 °F, the particulate filter shall show no ignition or structural damage, and shall pass airflow resistance (3.3.1) and filtration efficiency (3.3.2) requirements.

3.4.2 High humidity. After conditioning at 80% relative humidity or above and 113 °F for 7 days, the particulate filter shall pass airflow resistance (3.3.1) and filtration efficiency (3.3.2) requirements.

3.4.3 Structural integrity.

3.4.3.1 Structural deformation. The particulate filter shall be capable of withstanding repeated applications of a quasi−static mechanical load of 12900 newtons (2900 pounds) in the axial direction. Each gasket shall not become dislodged from the filter by the process of loading a filter into a housing and removing it. After testing, the filter shall meet the filtration efficiency requirement in 3.3.2.

3.4.3.2 Shock. After shock treatment, the particulate filter shall pass airflow resistance (3.3.1) and the filtration efficiency (3.3.2) requirements.

3.4.3.3 Vibration. After sinusoidal vibration between 4 and 50 Hz, the particulate filter shall pass airflow resistance (3.3.1) and the filtration efficiency (3.3.2) requirements.

3.4.3.4 Rough handling. After rough handling, consisting of two hundred 3/4−inch drops per minute for 15 minutes, the particulate filter shall show no evidence of cracks or dents, and shall pass airflow resistance (3.3.1) and filtration efficiency (3.3.2) requirements.

3.4.4 Fungus. The materials of construction, except for ASZM−TEDA carbon, shall not support fungus and mildew growth beyond trace levels.

3.4.5 Corrosion. Any construction material shall resist salt fog corrosion to an extent equal or greater than anodized aluminum when the aluminum alloy number is in the 3000 to 5000 series of ASTM B209.

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3.4.6 Dust capacity. The particulate filter shall deliver the rated airflow at a resistance of no greater than 5.7 inch of water when uniformly loaded with 2 kilograms (4.4 pounds) of dust per Class A2 of ISO 12103−1.

3.4.7 Overpressure. The particulate filter shall satisfy the minimum efficiency requirement of 3.3.2 after being subjected to a repeated overpressure that is produced by an airflow of 400 scfm.

3.4.8 Flammability. The particulate filter, when clean, shall not produce flame, sparks or smoke in quantities greater than that allowed for a Class 2 air filter unit per UL 900.

3.5 Ownership and support

3.5.1 Shelf life. Filter shelf life shall be at least 5 years (extendable) (see 6.6). Shelf life is the maximum time an item may be in sheltered storage within its packaging and still remain fit for use.

3.5.2 Safety, health, environment.

3.5.2.1 Materials. The materials of construction shall be non−toxic, non−hazardous, and non−carcinogenic. They shall not be capable of having any adverse effects on human health during normal use. If special use conditions create a health hazard potential, the equipment shall have appropriate hazard warning labels in accordance with 29 CFR 1910.1200.

3.5.2.2 Particulate emissions. The filter shall pose no potential inhalation hazard to the user, and whose normal use will not exceed the lower of (a) 8−hour time weighted average Threshold Limit Values (TLVs) published by the American Conference of Governmental Industrial Hygienists (ACGIH), or (b) 8−hour time weighted average Permissible Exposure Limits (PELs) regulated by Occupational Safety and Health Administration (OSHA) and found in Material Safety Data Sheets (MSDS).

3.5.2.3 Odors. When air is forced through the filter at 200 scfm, it shall not emit an irritating level of particulate emissions or obnoxious odors.

3.5.2.4 Hazardous waste. At the time of disposal, uncontaminated filters (including any coatings) shall not be a Resource Conservation Recovery Act hazardous waste according to characteristics cited in 40 CFR 261.21−261.24 and shall not be listed as specific hazardous waste chemicals in 40 CFR 261.33(e) and (f).

3.5.2.5 Sharp edges. Filters shall be free from burrs and sharp edges that might injure personnel, damage filter media, or damage Mission Oriented Protective Posture (MOPP) IV gloves during handling.

3.5.3 Identification markings. All markings shall be accessible, legible and permanent.

(a) The filter shall contain markings on the outside surface which include the name “FILTER, PARTICULATE, 200 CFM”, contract number, lot number, serial number, national stock number, cage code, manufacturer’s name, and manufacture date. Letters shall be at least 0.2 inch high.

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(b) The filter shall also contain a barcode from which the following may be read: name, contract number, lot number, national stock number, cage code, manufacturer's part number, serial number, manufacture date, expiration date, and measured airflow resistance and filtration efficiency. Barcode dimensions and characteristics shall comply with MIL−STD−130.

3.5.4 Workmanship

3.5.4.1 The filters shall be free from foreign matter (dirt, oil, or viscous materials).

3.5.4.2 The filter units shall be free from cracked, bent, or dented metal sections, and abraded gaskets.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2)
(b) Conformance inspection (see 4.3)

4.2 First article inspection.

4.2.1 Sample. The first article (FA) sample shall consist of 20 particulate filters unless otherwise specified in the contract. The FA sample shall be produced using the same methods, materials, equipment, and processes as will be used during regular production.

4.2.2 Inspections to be performed. Unless otherwise specified by the terms of the contract, the sample first article items shall be subjected to all of the verifications specified herein. Verification of requirements shall be by examination, (examination includes visual inspection with or without simple commercial measuring devices), demonstration, test data, or by Certificate of Compliance (COC) with analysis or supporting evidence. The Government reserves the right to accept or reject each COC. First article filters shall be inspected according to Figure 2, or as determined by the Government (see 6.2). Each filter shall be examined for interface characteristics, identification markings, workmanship, and safety hazards. Each filter shall then be tested for airflow resistance and filtration efficiency. Filters shall then be packaged using the same methods, materials, equipment, and processes as will be used during regular production. Packaged filters shall then be forwarded to the Government for rough handling and additional tests. Requirements and verification methods are cross referenced in Table I.

4.2.3 Acceptance criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

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TABLE I. First article inspection

Description	Requirement Paragraph	Verification Paragraph
Interface requirements	3.2	4.4.1
Size and shape	3.2.1	4.4.1.1
Weight	3.2.2	4.4.1.2
Interchangeability	3.2.3	4.4.1.3
Operating requirements	3.3	4.4.2
Airflow resistance	3.3.1	4.4.2.1
Filtration efficiency	3.3.2	4.4.2.2
Environmental requirements	3.4	4.4.3
Hot/cold temperature	3.4.1	4.4.3.1
High humidity	3.4.2	4.4.3.2
Structural integrity	3.4.3	4.4.3.3
Structural deformation	3.4.3.1	4.4.3.3.1
Shock	3.4.3.2	4.4.3.3.2
Vibration	3.4.3.3	4.4.3.3.3
Rough handling	3.4.3.4	4.4.3.3.4
Fungus and mildew	3.4.4	4.4.3.4
Corrosion	3.4.5	4.4.3.5
Dust capacity	3.4.6	4.4.3.6
Overpressure	3.4.7	4.4.3.7
Flammability	3.4.8	4.4.3.8
Ownership and support requirements	3.5	4.4.4
Shelf life	3.5.1	4.4.4.1
Safety, health, and environment	3.5.2	4.4.4.2
Identification markings	3.5.3	4.4.4.3
Workmanship	3.5.4	4.4.4.4

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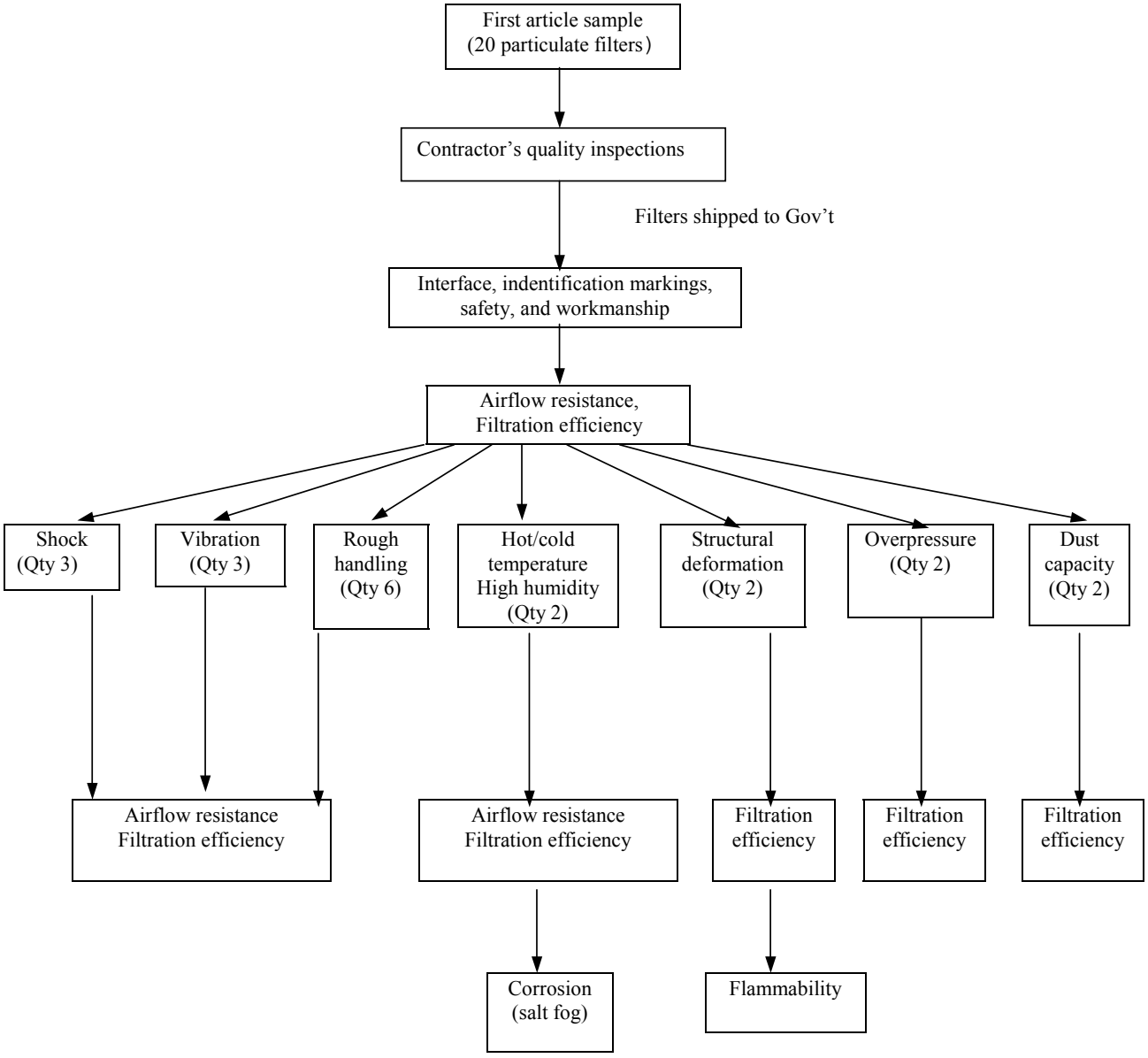


FIGURE 2. First article inspection sequence

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4.3 Conformance inspection.

4.3.1 Lotting. A lot shall consist of particulate filters produced by one manufacturer at one plant, from the same materials, by the same process, and without a break in production of more than 10 calendar days. Each lot shall be identified and controlled in accordance with MIL−STD−1168.

4.3.2 Sampling. Inspection and test of random samples shall be conducted in accordance with the classification of characteristics in 4.3.5, and when specified in Table II.

4.3.3 Inspection procedure. Every item in the lot shall be inspected for critical characteristics. Sample filters shall be examined and tested in accordance with the classification of characteristics in 4.3.5. Failure of any sample to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented (see footnote, Table II). Any filter samples forwarded to the Government for testing shall be packaged such they will be protected from moisture in the air. One or more environmental tests shall also be conducted if there is a materials change that increases the risk of failure for a particular environment.

4.3.4 Inspection characteristics. Critical characteristics are those whose nonconformance to specified requirements is likely to result in hazardous or unsafe conditions for individuals using, maintaining, or depending upon the product. Also those characteristics whose nonconformance to specified requirements is likely to prevent performance of the tactical function of a major end item. Major characteristics are those whose nonconformance to specified requirements is likely to result in failure or to reduce materially the usability of the item for its intended purpose. Minor characteristics are those whose nonconformance to specified requirements is not likely to reduce materially the operation or usability of the item for its intended purpose.

4.3.5 Classification of characteristics. Conformance examinations and tests shall be as specified in Table III. When specified herein, accept on “0” and reject on “1” attributes sampling inspection shall be performed on the designated characteristics using the stated levels in Table II for selection of sample sizes.

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TABLE II. Sampling

	Inspection level and samples sizes										
Lot size	I	II	III	IV	V	VI	VII	VIII	IX	X	XI
2 to 8	*	*	*	*	*	*	*	*	5	3	2
9 to 15	*	*	*	*	*	*	13	8	5	3	2
16 to 25	*	*	*	*	*	20	13	8	5	3	3
26 to 50	*	*	*	*	32	20	13	8	5	5	5
51 to 90	*	*	*	50	32	20	13	8	7	6	5
91 to 150	*	*	125	50	32	20	13	12	11	7	6
151 to 280	*	*	125	50	32	20	20	19	13	10	7
281 to 500	*	315	125	50	48	47	29	21	16	11	9
501 to 1200	*	315	125	75	73	47	34	27	19	15	11

* Asterisk means 100% inspection of lot. Accept the lot if there are zero non-conforming characteristics and reject the lot if there are one or more non-conforming characteristics for all inspection levels.

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TABLE III. Classification of characteristics

PARAGRAPH 4.3.5	TITLE Filter, Particulate		Sheet 1 of 1	DRWG NO.
CATEGORY	**CHARACTERISTIC**	**SAMPLING**	**REQUIRE-MENT PARAGRAPH**	**INSPECTION METHOD**
Critical	Filtration efficiency	100%	3.3.2	4.4.2.2
Major				
101	Airflow resistance	100%	3.3.1	4.4.2.1
102	Over- pressure and filtration efficiency*	Table II, level IX	3.4.7 3.3.2	4.4.3.7 4.4.2.2
103	Safety, environment & health	Table II, level IX	3.5.2	4.4.4.2
Minor				
201	Identification markings	Table II, level IX	3.5.3	4.4.4.3
202	Workmanship	Table II, level IX	3.5.4	4.4.4.4

NOTES:

A- Analysis
D – Demonstration
E - Examination
COC – Certificate of Compliance

*Filtration efficiency is categorized as a critical defect

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4.4 Verification methods and procedures.

4.4.1 Interface verification.

4.4.1.1 Size and shape. Verify by examination and measurement using commercial inspection equipment.

4.4.1.2 Weight. Verify by examination and measurement using commercial inspection equipment.

4.4.1.3 Interchangeability. Verify by certification and analysis of filter design and surrounding interfacing hardware.

4.4.2 Operating verification.

4.4.2.1 Airflow resistance. The airflow resistance shall be determined at the rated airflow of 200 scfm. (Air shall flow through the filter radially outward.) The measured pressure drop across the filter, when corrected to standard conditions of 70˚F and 1 atm (760 mm Hg), shall be no greater than that specified in 3.3.1. The up−stream and down−stream static pressure measuring tubes shall be as close as possible to the filter and shall not be on a section of duct that has a changing cross sectional area. Test for airflow resistance as follows:

(a) Connect the filter to a source of forced air.
(b) Set the flow of air through the filter to 200±10 cfm.
(c) Record the barometric pressure.
(d) Measure and record the air stream temperature. This temperature should be no higher than 125˚F.
(e) Determine and record the difference between up-stream and downstream static pressure.

If the recorded air stream temperature and barometric pressure is different than 70˚F and 1 atm (760 mm Hg), calculate and record the air flow resistance (ΔP (cal)) using the equation below to correct the measured airflow resistance to airflow resistance at standard conditions:

$$\Delta P\ (cal) = \Delta P\ (measured) \times P\ (test) \times 86.21 / [(460 + T)^{1.768}]$$

where:

ΔP (cal) – air flow resistance corrected to standard conditions in inches of water gage (iwg)
P (test) – barometric pressure at time of test in millimeters of mercury (mm Hg)
ΔP (measured) – air flow resistance from test measurement in iwg
T – temperature of air stream flowing through the filter in degrees F

Note: Correction for nonstandard conditions is not required when it is clear that the passage or failure of the filter is not in doubt.

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4.4.2.2 Filtration efficiency.

4.4.2.2.1 Methodology. Filtration efficiency shall be measured for each sample particulate filter by using Method A or Method B using aerosolized polyalphaolefin as the simulant challenge aerosol (see 6.5). In Method A, the filter shall be challenged with a polydispersed aerosol whose count mean diameter is 0.3 microns and whose particle size standard deviation is no greater than 2.0. Method A shall use a particle analyzer for size and count determinations. Mass mean particle size should be measured using one first article filter for comparison with the count mean particle size. In Method B, the filter shall be challenged with a monodispersed or near monodispersed aerosol whose mass mean diameter is 0.3 microns and whose particle size standard deviation is no greater than 1.3. Method B shall use a photometer to measure aerosol concentrations. If particle size test data are log normally distributed, than geometric mean and standard deviation should be computed instead of arithmetic mean and standard deviation.

4.4.2.2.2 Apparatus. For either method, the test apparatus shall be capable of maintaining a stable concentration of aerosol with a steady flow through the filter of 200 scfm, accurately determining the penetration percent within ± 0.005%, and producing no adverse effects on human health. The aerosol selected should not react with any of the construction materials. The measurement equipment must have an indicator that alerts the user when the equipment is in need of calibration or maintenance. The downstream sample point shall be located such that changing of its location across the ductwork does not cause a significant change in the measured concentrations. This shall be verified using a filter with a known measurable leak.

4.4.2.2.3 Procedure. The particulate filter shall be placed in the test apparatus conditioned at room temperature. Maintain the airflow within 10% of 200 scfm with air flowing radially outward through filter. That airflow shall be maintained while filtration efficiency is recorded. Challenge the filter at a particle count concentration (Method A) or aerosol concentration (Method B) that is within the acceptable range of the equipment manufacturer's specification. If using Method B, challenge the filter at an aerosol concentration of 100±20 μg/L. An inlet concentration lower than 100±20 μg/L may be used during conformance testing if the range is within the manufacturer's specification. Filtration efficiency shall be calculated from the particle counts or concentrations as follows:

Penetration % = N_d/N_u * 100 for Method A
Penetration % = C_d/C_u * 100 for Method B

Filtration efficiency % = 100 – penetration %, where

C_d = Concentration downstream of the filter
C_u = Concentration upstream of the filter
N_d = Number of particles downstream of the filter
N_u = Number of particles upstream of the filter

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4.4.3 Environmental verification.

4.4.3.1 Hot/cold temperature. Condition two filters for 8 hours at −50˚F in an environmental chamber. Remove filters from chamber and examine them for damage at room temperature. When temperature of filters returns to 70 ±10 ˚F, check airflow resistance and particulate filtration efficiency. Then, condition two filters for one week at 160˚F in an environmental chamber. Remove filters from chamber and examine them for damage at room temperature. When temperature of filters returns to 70 ±10 ˚F, check airflow resistance and particulate filtration efficiency.

4.4.3.2 High humidity. Expose the particulate filter to 80% relative humidity at 84˚F for 7 days. After removing filter from environmental chamber, perform airflow resistance and efficiency testing.

4.4.3.3 Structural integrity.

4.4.3.3.1 Structural deformation. The particulate filter shall be subjected to a quasi−static mechanical load of 2900 pounds in the axial direction for 3 cycles of one minute per cycle. Retest filters for filtration efficiency.

4.4.3.3.2 Shock. Shock testing shall be performed on particulate filters when fully loaded as gas−particulate filter sets in the U.S. Navy CPS Filter System (53711−6263500). The CPS system has a flow capacity of 600 cfm. Filters shall be installed in accordance with the operation and maintenance manual, SS200−AG−MMM−010. The filter housing shall be filled to capacity. The filter system (housing with gas−particulate filter sets) shall be shock tested per MIL−S−901 using an approved mounting fixture. Shock test procedures shall be based on Class I (no resilient mountings), Type B (subsidiary component), and medium weight category defined in MIL−S−901. After shock testing, particulate filters shall be removed from the housing and individually retested for airflow resistance and filtration efficiency.

4.4.3.3.3 Vibration. Vibration testing shall be performed on particulate filters when fully loaded as gas−particulate sets in the U.S. Navy CPS Filter System (53711−6263500). The CPS system has a flow capacity of 600 cfm. Filters shall be installed in accordance with the operation and maintenance manual, SS200−AG−MMM−010. The filter housing shall be filled to capacity. The filter system (housing with gas−particulate filter sets) shall be vibration tested per MIL−STD−167−1 using an approved mounting fixture. Test procedure shall be based on MIL−STD−167−1, Type I− Environmental Vibration, 4 Hz to 50 Hz, and consist of 3 parts : (1) Exploratory test to determine resonances (2) Variable frequency test (4 to 50 Hz with 5 minute dwell time per hertz), and (3) Two−hour endurance test for each of the 3 orthogonal axes. After vibration testing, particulate filters shall be removed from the housing and individually tested for airflow resistance and filtration efficiency.

4.4.3.3.4 Rough handling. Perform rough handling testing on particulate filters that have not previously undergone shock and vibration testing. All sample filters shall be mounted with the cylindrical axis in a vertical orientation and parallel to the direction of movement. No filter shall be allowed to touch any other filter. Filters shall be rough handled in accordance with Method 105.11 of MIL−STD−282. The rough handling shall consist of 200 3/4−inch drops

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per minute for 15 minutes. Following rough handling, examine bottom plate and surrounding area for evidence of loose dust, and test for airflow resistance and filtration efficiency.

4.4.3.4 Fungus. Verify, if possible, by analysis of technical documentation such as specifications, design data, existing test reports, and operating experience. Otherwise, verify requirement by performing a fungus test in accordance with Method 508.5 of MIL−STD−810.

4.4.3.5 Corrosion. The manufacturer shall provide objective evidence and certify that any metallic construction materials will resist salt fog corrosion. Objective evidence includes test data or past service life data. If objective evidence does not exist or is disapproved by the Government, perform salt fog testing in accordance with Method 509.3 of MIL−STD−810 except the total duration shall be 20 days corresponding to five 96−hour exposure cycles. Each 96−hour exposure cycle is defined as 48 hours of salt fog challenge followed by 48 hours of storage at ambient laboratory conditions with no salt fog challenge. After the five 96−hour exposure cycles, rinse the external surfaces with clear water if desired, dry, and inspect filter (inside and outside) for evidence of damage and corrosion. If there is evidence of corrosion, test each particulate filter for airflow resistance and filtration efficiency.

4.4.3.6 Dust capacity. Prior to conducting the dust capacity test, the filter shall be conditioned for a period of at least 24 hr at ambient laboratory temperature and not more than 50% relative humidity. The dust capacity test apparatus shall conform to SAE J726. The apparatus shall be capable of dispersing a dust per grade A2 of ISO 12103−1. Provisions shall be made to determine the total weight of dust injected and for measuring the filter inlet pressure head. The preconditioned filter shall be placed in the test apparatus so that air flows radially outward through the filter during testing. The test shall consist of an initialization phase and a dust injection phase. During the initialization phase, flow through the filter will be adjusted to the rated flow without dust injection. During the dust injection phase, dust will be injected into the inlet air stream at the rate required to establish a dust concentration 1.0 $\pm$0.1 gm/m^3 at the rated airflow under ambient laboratory conditions. Record the inlet head values every 15 minutes after the initiation of the dust injection. The test shall be run until the inlet head reaches or exceeds the resistance value stated in 5.7 iwg, or for 6 hours, or until a mechanical failure is indicated, whichever occurs first. Upon completion of the required dust injection phase, record the total weight of dust injected, and test the filter for filtration efficiency. Any mechanical failure of the filter, such as a sudden surge of flow and/or drop in the inlet head pressure during the dust capacity test phase, or failure of the subsequent efficiency test shall be cause for rejection.

4.4.3.7 Overpressure. Subject the filter to a differential pressure produced by the airflow rate of 400$\pm$ 20 scfm. After the filter inlet is subjected to this overpressure for 5 minutes, the flow shall be stopped for one minute. This test cycle shall be repeated 3 times. After completion of the 4 overpressure cycles, perform efficiency testing at the rated 200 scfm airflow.

4.4.3.8 Flammability. The combustibility and smoke generating potential of the filter shall be determined by using the apparatus and procedures specified in section 5 and 6 of UL 900. The quantity of flame, sparks and smoke should be evaluated in accordance with procedures in UL 900.

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4.4.4 Ownership and support verification.

4.4.4.1 Shelf life. Verify shelf life by analysis (engineering evaluation) of materials of construction, packaging, and surveillance data if available. Assume storage temperatures up to 160˚F and relative humidities up to 100%.

4.4.4.2 Safety, health, and environment.

4.4.4.2.1 Materials. Verify no adverse effects on health by health hazard assessment or analysis. In the assessment, consider the probability and consequences of failing to meet each requirement.

4.4.4.2.2 Odors. Verify by COC based on existing human factors data collected on filters of the same design that are used in shelters.

4.4.4.2.3 Hazardous waste. Verify by certification based on analysis of construction materials and coatings. Certify that all construction materials including any coatings are free of hazardous waste as defined by the the Code of Federal Register.

4.4.4.2.4 Sharp edges. Verify by examination of edges on external surfaces.

4.4.4.3 Identification markings. Verify by examination.

4.4.4.4 Workmanship. Verify by examination.

5. PACKAGING

5.1 Packaging. For acquisition purposes, packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DOD personnel or in−house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD−ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The filter, particulate, 200 cfm is intended for use as part of a Nuclear Biological Chemical (NBC) air handling system of a collective protection system. The filter may be used in the shipboard CPS, in land−based CPE such as the Modular CPE and Simplified CPE, and in the transportable CPS. Filters are normally mounted with the axis in the horizontal position.

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6.2 Acquisition requirements. Acquisition documents should specific the following:

(a) Title, number, and date of this specification.
(b) First article:
  (1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Govern ment.
  (2) Name and address of test facility and shipping instructions when testing is performed by the Government (see 6.3).
  (3) Time required for the Government to notify the contractor whether or not to proceed with production.
(c) Packaging requirements.
(d) Data requirements (e.g. health hazard or risk assessment reports)
(e) Marking requirements if different than in section 3.
(f) Whether any Government Furnished Equipment is supplied.
(g) Label requirements. Adhesive−backed labels may need to be purchased so that user may record filter installation date.

6.3 Test facilities. Suggested shipping addresses for Government portion of First Article test are as follows:

(a) For navy−type shock and vibration testing:
NU Laboratories, Inc.
312 Old Allerton Road
Annandale, NJ 08801
(908) 713−9300

(b) For receipt, examination, and distribution to test activities at ECBC:
U.S. Army ECBC
AMSRD−ECB−ENE−M
Building E5165
APG, MD 21010−5424

6.4 Technical data. Information on one particulate filter is shown in Drawing 6573651 (Navy) and Drawing 5−19−6262 (Army). Filter cover with groove is shown in subsidiary Drawing 6573654. NSN is 4240−01−066−3266.

6.4.1 Filter media. Filter media should have a tensile strength of at least 866 gm/cm (4.85 lb/in) in the machine direction and at least 433 gm/cm (2.42 lb/in) in the cross direction.

6.4.2 Liners. Liners should be used on the inner and outer diameters of the filter media to carry axial loads and protect the media from damage.

6.4.3 Sealant charcteristics. Desirable characteristics of the sealant for securing filtration media to covers are as follows:

(a) Tensile strength of at least 91 kg/cm^2

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(b) Tear strength of at least 28 kg/cm^2.
(c) Adhesion within 5 minutes at 176˚C (350˚F)

6.5 Aerosol information. Candidate replacements aerosol materials for potentially hazardous dioctyl phthalate in penetrometer test machines are identified in report CRDEC−TR−271 which is available from Edgewood Chemical Biological Center, AMSRD−ECB−ENE−M, Bldg. E5165, Aberdeen Proving Ground, Maryland 21010−5424. The current type of aerosol used by Edgewood Chemical Biological Center is Durasyn® 164 polyalphaolefin available from Amoco Chemicals, 801 Warrenville Rd, MC 6018, P.O. Box 5206, Lisle, Illinois 60532.

6.6 Shelf life. This specification covers items where shelf life is a consideration. Specific shelf life requirements should be specified in the contract or purchase order. The shelf life codes are contained in the Federal Logistics Information System Total Item Record. Additive information for shelf life management may be obtained from DoD 4140.27−M; Shelflife Management Manual, or the designated shelf life Points of Contact (POC). The POC should be contacted in the following order: (I) the Inventory Control Points (ICPs), and (2) the DoD Service and Agency administrators for the DoD Shelf Life Program. Appropriate POCs for the DoD Shelf Life Program can be contacted through the DoD Shelf Life Management website: htm:/ /www.shelflife.hg.dla.mil/ .

6.7 Useful life. A useful life goal for the filter is 4 years or more of peace time use. The useful life is the maximum period of time an unpackaged item can remain fit for use.

6.8 Subject term (key word) listing.

Collective protection
Filtered air
HEPA filter

Custodian:

Army − EA
Navy − SH
Air Force − 99

Review activities:
Army − MR
DLA − IS

Preparing activity:

Army − EA
(Project No. 4240−2006−003)

Note: The activities listed above were interested in this document as of the date of this document. Since organizations change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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